

PARTNER BRIEFING

Halqa

Digital infrastructure for Pakistan's committee economy.

Presented by Taha Kayani

Structure: 1 The committee economy · 2 Failure modes · 3 Halqa: the three roles · 4 Earning engines and mathematics · 5 Turn market · 6 Vault and investments · 7 Treasury and custody · 8 Default protection · 9 Competitive landscape · 10 Application and technology · 11 Business model · 12 Demands · 13 Build inventory and delivery status

BUILT

operational today

LICENCE

requires SECP authorisation

LICENCE+BANK

requires authorisation and partner bank

The instrument

Mechanics (ROSCA — rotating savings and credit association)

- Fixed group of n members; identical installment each period.
- One member per period collects the full pool; n periods complete the cycle.
- Early positions receive interest-free credit, repaid in installments.
- Late positions hold enforced, disciplined savings.
- Net transfer over the full cycle is zero; the traded quantity is time.

The dual function

Each member is simultaneously lender and borrower. Members fund each other's turns and accept 0% because each receives a 0% turn in return. No intermediary spread exists — which is why the pricing is sustainable, and why an institution inserting itself must break it.

Foundations

FOUNDATION	FUNCTION
Social collateral	Group-embedded enforcement outperforms legal paperwork at this scale (Besley, Coate & Louny, AER 1993).
Riba-free structure	No interest instrument anywhere in the cycle.
Zero overhead	No branches, forms, or minimums; operating cost near zero.
Commitment mechanism	The rigidity is the product: members join in order to be bound to save.

Why banks hold neither side

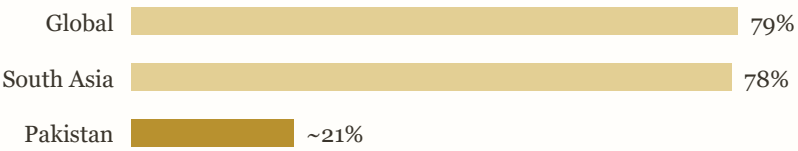
The credit side runs on social information banks cannot observe; the 0% price requires the absence of a funding spread banks cannot escape. Banks captured only the savings shell (recurring deposit schemes), which loses to the kameti because the rotation is the demanded product.

One structure worldwide, and the Pakistani figures

MARKET	FORM	SCALE REFERENCE
Pakistan	kameti / BC	41% adult participation; 25–30M active
Egypt	gam'eya	Money Fellows: 8.5M+ users, profitable
India	chit funds	Regulated (Chit Funds Act 1982); licensed foremen; capped commissions; operators at multi-billion scale
South Africa	stokvels	11M members; 800k+ groups
Indonesia	arisan	Nationally ubiquitous; platform-integrated (Mapan)
Mexico / W. Africa / Korea	tanda / susu / kye	Documented in development-finance literature

The identical structure recurs independently across unconnected cultures — the mechanism is economic, not folkloric.

Account ownership, adults (Findex 2025)



Annual committee flow, Pakistan (Rupees)



All monetary figures in rupees. Pakistan is the regional inclusion outlier by nearly four times; the committee flow is of national-budget magnitude and entirely unrecorded.

One of eight countries holding over half the world's 1.3B unbanked adults. 240M population → ~140M adults → ~57M ever participated → 25–30M active today.

How the instrument fails

Case: Sidra Humaid, Karachi, 2022

- Committees operated across **117 WhatsApp groups**.
- Collections taken into **her personal custody**.
- **Rs 420 million** collected; no written records maintained.
- No repayment capacity; declared inability to pay.
- Member recovery: **zero**. Without records, victims had no legal existence as creditors.

The structural pattern

Self-custody plus absence of records equals unrecoverable loss. The organiser holds both the money and the only memory of who is owed what.

Frequency

- Large-scale precedents: Double Shah (2005–07); modaraba scandals (2013–14).
- Small-circle collapses occur continuously and go unreported — a defaulted street committee produces no filing, no record, no statistic.
- Each unrecorded collapse pushes savers toward smaller, poorer circles; the failure mode taxes the entire system's scale.

The diagnosis

The committee's weakness is not its economics — those have survived a century on every continent. The weakness is the absence of **records, collateral, scoring, and recovery**.

That absence is the opportunity.

Why members prefer committees over bank products

REASON	MECHANISM	WHY THE BANK CANNOT MATCH IT
Interest-free	No riba anywhere in the cycle; a binding religious constraint for a large share of the population.	Bank deposits and loans are interest instruments by construction.
Credit access	The circle approves on years of direct social observation.	Banks require documented income, collateral, and history — which this population lacks. The circle's information is non-transferable to an institution.
Price	Early liquidity at 0%, funded mutually by members who each receive their own 0% turn.	Bank funds carry deposit cost, branch cost, and capital charges; consumer credit prices at 25–40%. A bank 0% product loses money per transaction.
Zero friction	No forms, branches, or minimum balances.	Onboarding and servicing costs exceed lifetime revenue at these tickets.
Commitment	Social pressure enforces saving; the rigidity is the purchased product.	A deposit account is voluntary and frictionless to empty — the opposite of what the saver is buying.

Part 1 conclusion: demand proven at national scale; failure mode precisely identified; incumbent institutions structurally unable to serve it directly.

Definition and the four roles

Halqa is the infrastructure layer for committees. Members continue transacting with each other; the platform operates the system around them. Halqa is never the borrower, never the lender, never the counterparty. Its four roles:

ROLE	FUNCTIONS	STATUS
1 · The Ledger	Double-entry record of every obligation, payment, deposit, and payout in integer paisa · gives members legal existence as creditors · append-only audit trail.	BUILT
2 · The Conductor	Runs and enforces the committee: schedules rounds, sequences turns, holds payment deadlines, applies holdbacks and progressive penalties, resolves defaults through the recovery waterfall, and administers the rehabilitation path · the automated referee that never forgets or plays favourites.	BUILT
3 · The Treasury	Custody of collections in segregated bank escrow · placement of idle balances into approved instruments (money-market, income funds, gold-linked, equity ETFs) · netting, liquidity ladder, buffers, daily reconciliation · payout execution against the calendar.	LICENCE+BANK
4 · The Credit Bureau	Reliability scoring updated by every payment · credit-weighted turn ordering · exportable credit passport per member · host reputation records · reporting rail into eCIB, DataCheck, Tasdeeq.	BUILT (bureau reporting LICENCE)

Division of function: members remain each other's credit source — preserving the 0% economics. The partner bank provides account custody, fund management (Mudarib), and payment rails. Halqa operates the ledger, the treasury logic, and the credit bureau between them.

The composable engine — five families of levers

The profit engine is a set of independent levers grouped into five families. The host switches any combination on at creation; none is mandatory; an unconfigured circle is a plain rotation. Every active lever is disclosed to members before joining.

FAMILY	LEVERS AND FUNCTION	STATUS
A • Yield generate the profit pool	Float placement — idle pool placed in an Islamic money-market fund. Deposit remuneration — security deposits earn on the same basis. Group allocation — a host-set slice of the pool invested into a dated scheme.	L+B
B • Distribution split the pool fairly	Capital-days basis — profit split by amount × days committed. Patience weighting — weight rises 1.0×→2.0× by position. Pooled dividend — a round dividend shared evenly rather than to one position.	BUILT
C • Behavioural reward conduct	Early-payment boost — 1.25× weight for consistently early payers. Clean-streak bonus — +5% float profit per consecutive clean round, capped +50%.	BUILT
D • Liquidity pricing price time	Early-access fee — position 1 pays 10% of pool declining to zero; 90% to waiting members, 10% platform.	BUILT
E • Compounding extend the earning period	Delayed distribution — a settle-window before release, lengthening earning time. Profit recycling — realised profit re-committed into the next round's float.	BUILT

Families A and E create and extend the pool; B, C and D govern how it is shared and how early liquidity is priced. A previously available prize-draw lever was removed on Shariah grounds.

The mathematics — every lever, named

Family A — Yield

Float placement / Deposit remuneration

$$\text{profit} = \text{principal} \times \text{rate} \times \text{days} / 365$$

Group allocation

$$\text{allocatedCapitalDays} = \text{capitalDays} \times \text{allocationBps} / 10000$$

$$\text{productive} = \text{allocated} \times (10000 - \text{reserveBps}) / 10000$$

Family B — Distribution

Capital-days basis (the fairness measure)

$$\text{capitalDays}_i = \Sigma_{\text{round}} \text{contribution} \times n \times \text{daysInvested}$$

$$\text{share}_i = \text{weight}_i / \Sigma \text{weight}$$

Patience weighting

$$w(k) = 1 + (k-1)/(n-1) \quad [1.0 \times \dots 2.0 \times]$$

Pooled dividend

$$\text{dividend}_i = \text{roundDividend} / n$$

Family C — Behavioural

Early-payment boost

$$\text{boost} = 1.25 \text{ if } \text{earlyCount} \times 4 \geq \text{paidCount} \times 3$$

Clean-streak bonus

$$\text{streak} = \min(0.05 \times \text{cleanRounds}, 0.50)$$

Family D — Liquidity pricing

Early-access fee

$$\text{fee}(k) = 0.10 \times (n-k)/(n-1) \quad [90\% \rightarrow \text{members}]$$

Family E — Compounding

Delayed distribution / Profit recycling

$$\text{days} += \text{delayDays} ; \text{nextPool} += \text{realisedProfit}$$

Combined member profit

$$\text{weight}_i = \text{capitalDays}_i \times w(k_i) \times \text{boost}_i$$

$$\text{memberProfit}_i = \text{pool} \times (1 + \text{streak}) \times \text{share}_i$$

All arithmetic in integer paisa; every formula locked as an automated test.

Estimated returns by deployment option

Indicative annual rate by deployment instrument (dated)



Rates are indicative and dated; equity and gold move with the market.

Best-member annual figure by configuration (Rs)

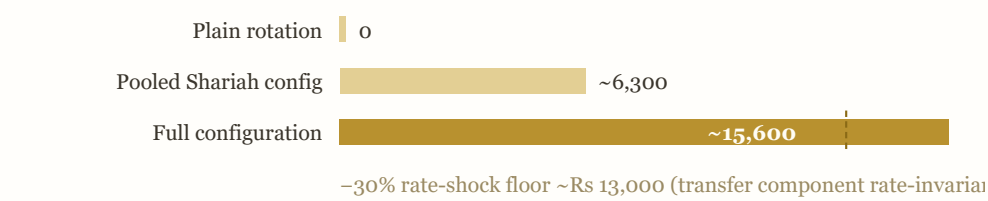


Illustration on the reference pool

Reference circle: 12 members, Rs 11,667/month, pool Rs 140,004, 70% coverage. One round's idle pool placed for ~24 days:

INSTRUMENT	INDICATIVE	~24-DAY PLACEMENT
Money-market	10.8%	~Rs 994
Income fund	12.1%	~Rs 1,114
Gold-linked	~15.0%	~Rs 1,381

Per-round float profit on the pool, before the distribution levers allocate it across members. Deposits earn in parallel over the full cycle, which is why the composed best-member annual figure reaches ~Rs 15,600.

Optionality is the design: the same platform serves a pure record-keeping rotation and a fully-configured circle, riba-free in both.

Buying and selling turns at a premium BUILT

Mechanics

- A member lists an upcoming payout position at an asking premium.
- Constraint: $0 \leq \text{premium} \leq 50\%$ of the payout — desperation pricing structurally excluded.
- Bids must exceed the standing floor (highest open bid, else the ask).
- Settlement is atomic: position swap, premium transfer, and platform fee in one transaction.
- The buyer pays the seller directly — value remains inside the circle.
- Platform fee: **10% of the premium.**

Function and constraints

- Prices the committee's time-value openly, replacing informal position-swapping by favour or pressure.
- Implied cost of acceleration is a fraction of informal lending rates; the counterparty is a fellow member, not a lender.
- Reliability scores unaffected — trading a position is a preference, not a breach.
- Group visibility and consent are structural; no trade occurs privately.
- **Labelling rule:** circles carrying Shariah-compliant engine labels permit zero-premium swaps only; premium trading operates in configurations that carry no such label.

The investment layer BUILT (live deployment LICENCE+BANK)

Two surfaces draw on the same instrument set. The **Vault** is the member's personal savings and investment account. The **Terminal** lists the dated-rate schemes into which both committee float and vault balances are placed. All placement is Mudarabah-based; the platform's share is 5% of profit on withdrawal, never principal.

Vault — personal facility

- Allocation across four sleeves in whole percentages summing to 100.
- Blended accrual rate = $\Sigma(\text{share}_i \times \text{rate}_i)$; blended risk score shown live.
- Payout parking, top-ups from Rs 100, full-sweep withdrawal.
- Growth projection: conservative / expected / optimistic paths, monthly compounding, asymmetric downside on the crypto sleeve.
- Per-sleeve risk score (/10), volatility range, exit liquidity in days.
- Money map: states the custody location of every rupee at each stage.

Terminal — scheme catalogue

- Dated indicative rates with source and as-of date on every scheme.
- Cadence-to-tenor filtering: only instruments that settle within the circle's period are offered.
- Committee float and security deposits placed here under the treasury logic (Part 7).
- The same catalogue backs both the vault sleeves and the circle engines.

Non-custodial position: Halqa never holds units or cash. Escrow is at the bank; units are at the fund trustee; Halqa instructs within member-signed mandates.

The four sleeves, explained

SLEEVE	INSTRUMENT AND MEANING	PURPOSE	LIQUIDITY AND RULE
Money-market	Units of an Islamic money-market fund (short government-backed ijarah and placements).	Capital stability; the default float sleeve. Rate resets with the market; principal does not swing.	T+1
Income / ETF	Islamic income funds (sukuk, screened instruments) and exchange-traded fund exposure — a listed basket of securities tracking an index, bought and sold as one unit.	Higher yield and market-linked growth for members accepting movement in exchange for return.	T+1 to T+2
Gold-linked	Fund units tracking the gold price. A hedge — a position held because it moves opposite to a risk already carried: when the rupee weakens, rupee gold typically rises.	Protects savings against rupee depreciation and inflation, the risk every saver already carries. It can fall in calm years — a hedge, not a guarantee.	~T+3 · permitted only where circle period ≥ 10 days, liquidation scheduled ahead of each payout
Crypto	A digital-asset basket. Explicitly non-Shariah and high-risk.	Offered only in the personal vault for members who insist; requires a recorded risk acknowledgement.	Variable · structurally excluded from committees — no circle can ever touch it

Auto-cover: a missed committee installment is settled automatically from the member's own vault balance before any penalty escalation — the vault doubles as the member's personal default buffer (Part 8).

Account architecture and collections

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Physical accounts

ACCOUNT	PURPOSE
Omnibus client-money escrow	Holds all member funds; trust status; members' beneficial ownership; bankruptcy-remote from Halqa.
Investment settlement	Cash legs of fund purchases and redemptions, separated from payout-ready funds.
Operating (Halqa)	Platform fees and late-payment revenue; strictly outside the client-money perimeter.

Custody position

Halqa instructs within member-signed mandates limited to named schemes and members' registered accounts. Client money never appears on Halqa's balance sheet. The deposit-taker is the licensed bank.

Collections — virtual IBAN per circle

Each committee receives its own virtual IBAN under the escrow. Pakistani IBAN format: **PK + 2 check digits + 4-character bank code + 16-digit account number** (24 characters). Inflows arrive pre-tagged by circle; reconciliation is automatic.

Payment rails

RAIL	USE
Raast	P2P disbursement · Request-to-Pay collection · Alias directory (pay-to-phone).
1Link	IBFT disbursement · 1Bill presentment: an installment payable in any bank application or over the counter.
Mobile wallets	Cash-in endpoints for the unbanked segment.

Deployment and controls

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Deployment path

- **Daily netting engine:** all circles' purchase and redemption demand aggregated into one order per fund per day — pooled execution, minimal transaction drag.
- Execution through **AMC institutional accounts** via order API, or **CDC Emlaak** (multi-AMC distribution platform: one integration, several fund houses).
- Fund units held by the fund trustee (**CDC**); cash by the bank; instructions by Halqa — three separated parties.
- **Per-circle unit ownership in the ledger:** pooled execution, segregated attribution — one circle's gold gain cannot leak into another; one circle's redemption cannot touch another's principal.
- Daily **NAV feed** supplies the dated rates displayed in-application.

Controls

CONTROL	RULE
Liquidity ladder	Instrument liquidity $\leq$ circle period – 7 days; nothing may be bought that cannot be sold before the payout it funds.
Buffers	15% payout buffer and 10% liquidity reserve held liquid regardless of placement.
Settlement discipline	Money-market T+1; gold T+3 with liquidation scheduled ahead of each payout.
Reconciliation	Daily diff of the bank statement feed (webhook or MT940/CAMT.053) against the ledger; any unexplained break freezes new placements.

Treasury position in one line: member money is either in escrow, in a short-settlement approved instrument, or en route to a payout — and every rupee is reconstructable from the ledger at any moment.

Admission controls and collateral BUILT

Pre-entry gates

CONTROL	MECHANISM
Risk-score ceiling	Host sets a target risk score; the circle's configuration is held within it.
Reliability gate	Admission requires standing; every payment updates the member score.
Exposure by standing	No fixed newcomer ceiling; a less-proven member's exposure is sized by score and a larger graduated deposit.
Credit-weighted ordering	Highest-scored members take the highest-exposure early turns — adverse-selection control.
Guarantor	Optional co-signer required to join, at host discretion.
Promissory note	Optional recorded undertaking captured at join.
Auto-debit mandate	Optional standing instruction against the member's account.

Collateral and reserves

CONTROL	MECHANISM
Graduated deposit	Deposit = remaining dues × coverage. Coverage = base (30–90%, default 70%) × (n–k)/(n–1), adjusted ±10–15 pts by credit standing, capped base+5%. Position 1 posts most; last posts zero.
Dynamic deposit	Deposit sizing recomputed as standing changes through the cycle.
Profit-as-collateral	Accrued profit held against obligations until the member is clear.
Payout buffer	15% of pool kept liquid regardless of placement.
Liquidity reserve	10% reserve against timing shocks.
Insurance reserve	Optional reserve provision per policy.
Safety Fund	Optional slot-fee pool (0.5–3%) guaranteeing a member's payout up to its balance; the circle's own money, remainder returned.

In-cycle enforcement and post-default resolution BUILT

In-cycle

CONTROL	MECHANISM
Payout holdback	A payout releases only when the member's obligations are current; configurable release after N clean payments.
Vault auto-cover	A missed installment settles from the member's own vault balance before escalation.
Smart nudges	Automated reminders ahead of due dates.
Peer nudges	Group-visible prompts engaging social pressure.
Anomaly monitoring	Payment-pattern surveillance across circles.
Delayed distribution	Profit held a set window, keeping members engaged to cycle end.

Post-default

CONTROL	MECHANISM
Progressive late fees	2% / 5% / 10% of installment by lateness tier — platform revenue ; reliability score deltas -10 / -20 / -40.
Feature lock	Platform features locked on default until cured.
Deposit forfeiture	Account lock; the defaulter's deposit forfeits into the recovery pool for affected members.
Recovery waterfall	Forfeited collateral and Safety Fund applied to make the group whole in priority order.
Post-receipt penalty	Additional score penalty where a member defaults after collecting.
Rehabilitation cooldown	Structured return path after a defined cooldown; default is recoverable-from.

Agent-based simulation of calendar 2025 BUILT

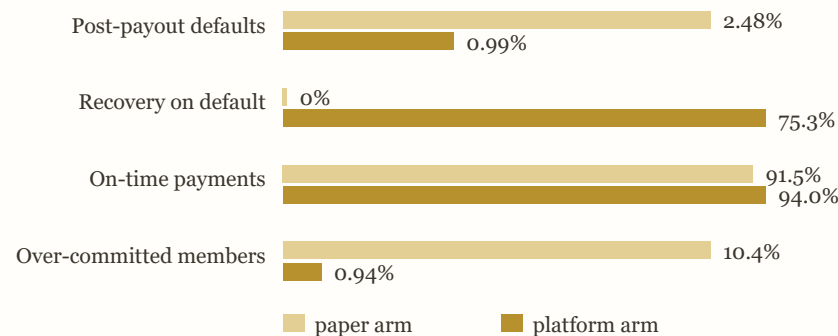
Method

- **120,000 agents · 10,441 committees** on the real 2025 calendar: Ramadan, both Eids, school-fee seasons, monsoon flooding.
- **Dual-arm, decision-paired randomness:** identical agents live the identical year twice — paper arm and platform arm — same incomes, shocks, and temptations. Only the platform differs; outcome differences are causal.
- **Deterministic seed:** every figure reproduces exactly on re-run; the simulation executes in continuous integration.

Seasonality

Both arms stress at Eid and peak during the floods — monthly default peak 376 (paper, July) against 173 (platform, August). The platform curve is lower in every month; the widest gap is early-cycle, where credit-weighted ordering concentrates reliable members in exposed turns.

Results — paper arm vs platform arm



Member engine yield across the simulated cohort: ~Rs 69.7M.

The field

PLAYER	MODEL	ASSESSMENT
Register applications Kameti.pk, iKAMETI, Circlys	Digital record books for existing circles.	No custody, no protection stack, no earning function, no credit output. Not at scale.
Oraan	600,000+ savers; SECP NBFC licence (2023). Pools strangers, takes custody, underwrites default risk on its own book, charges members tiered management fees — highest for early positions.	The category's credibility precedent. The management fee is the revenue; social collateral is replaced by company underwriting and a recovery team.
Halqa	Existing social circles retained; the composable engine returns float profit to members; the early-access fee redistributes 90% to members; default risk carried by collateral and the protection stack; channel economics — institutions pay for deposits and data.	A different structural bet: preserve the social layer, monetise the by-products.

International reference points

- **Money Fellows (Egypt)** — 8.5M+ users, profitable: the model at scale in a directly comparable economy.
- **India's chit funds** — the regulated end-state: licensed operators, capped commissions, statutory deposits.
- **Esusu (US)** — a billion-dollar valuation built on the informal-payment credit-data layer alone.

Member-facing features BUILT

FEATURE	FUNCTION
Committee lifecycle	Create, invite by code, join, pay, collect; goal-tagged circles (Hajj, fees, wedding, home).
Range	Periods 1–365 days; membership 3–150 (intervals ≤ 30 days above 30 members); daily circles run as plain rotations with engines disabled and the constraint stated.
In-committee chat and notices	Member-to-member communication and committee updates inside each circle.
Rafa assistant	Typo-tolerant retrieval (bounded edit-distance matching), romanised-Urdu synonym handling, conversational context, guided first-circle walkthrough.
Urdu interface	One-tap language switch with full right-to-left layout.
Real-time updates	Payment and payout events pushed over WebSocket; notification centre.
Credit passport	Exportable, verifiable repayment history per member.
Host reputation card	Public record of a host's completed circles and conduct.
Vault, terminal, turn market	Part 5 facilities integrated in the same application.
Distribution	Installable PWA; native Android and iOS builds (Capacitor).

Technology register BUILT

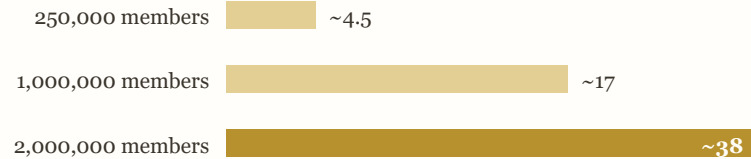
COMPONENT	FUNCTION	SIGNIFICANCE
TypeScript 5.8	Statically typed codebase, API and interface	Type errors in money paths caught at compile time.
Node.js / Express	API runtime and routing	One language across the stack.
PostgreSQL 17	ACID relational store	Transactional integrity for all financial state.
Prisma	Schema-as-code ORM	Versioned data model; typed queries; atomic multi-leg transactions.
Double-entry ledger (in-house)	Every movement one debit + credit in integer paisa (BigInt)	Money conserved by construction; no floating-point drift; balances derived.
Idempotency keys	Exactly-once money operations	A duplicated request cannot double-post a payment.
JWT + bcrypt	Sessions and credential hashing	Passwords never stored readable.
Zod	Runtime input validation	Every request checked against declared schemas before logic runs.
socket.io	WebSocket channel	Live payment and payout events without polling.
pino	Structured JSON logging	Production observability per request.
React 18 + Vite	Interface and build tooling	Code-split pages for low-end devices.
vitest • fast-check • supertest	Unit, property-based, and HTTP testing	Engine outputs locked as assertions; randomised edge-case search; 271 integration checks.
GitHub Actions	Continuous integration	Regressions cannot land silently.
Deterministic agent-based simulation	Seeded, reproducible population model	The validation instrument behind every protection claim.

Revenue register

LINE	RATE	PAYER	STATUS
Deposit-mobilisation commission	~1.25%/yr on average balances	Bank	L+B
Service fee	Rs 49/member/month; hosts free	Member	BUILT
Late-payment fees	2% / 5% / 10% of installment by tier	Late member	BUILT
Early-access fee share	10% of fee (90% to members)	Optional flow	BUILT
Turn-market fee	10% of premium	Optional flow	BUILT
Vault profit share	5% of profit; never principal	Optional flow	BUILT
Receivables origination + servicing	1–1.5% + 0.5%/yr	Institution	L+B
Fund distribution trail	~0.5%/yr on placed AUM	AMC	LICENCE
Takaful distribution	15–20% of contribution	Insurer	LICENCE
Credit-data services	Per inquiry	Institution	LICENCE

Majority institution-paid. The member's mandatory cost is the service fee; other member-side lines attach to optional flows or to breach of schedule.

Monthly revenue at scale (Rs crore)



Zero-opt-in floor at 250,000 members: ~Rs 2.4 crore/month — service fee plus commission on balances alone; no engine participation assumed.

Balance-sheet contribution

~Rs 45B in balances at 1M members — approximately 6% of the microfinance sector's deposit base; 15–25% deposit growth for a single institution; granular, scheduled, and pre-scored.

From the partner bank

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ITEM	PURPOSE
Escrow suite (4 accounts)	Client-money omnibus (trust status, bankruptcy-remote), investment settlement, charity/penalty routing, operating — segregated custody and clean audit.
Virtual IBAN range	One PK-format virtual IBAN per committee, unlimited issuance, zero per-IBAN fees, auto-tagged inflow reporting.
Statement / event feed	Intraday webhook of credits and debits, virtual IBAN on every line; fallback MT940/CAMT.053. Feeds daily reconciliation.
Cutoffs and settlement SLAs	Published daily cutoff for same-day fund orders; payout instructions executed within banking hours; written remedies.
Sandbox / UAT environment	Full test environment with fake money and a certification checklist before go-live.
Raast sponsorship	Access to Raast P2P (disbursement), Request-to-Pay (collection), and the Alias directory.
Auto-debit / standing instruction execution	Execution of digitally-captured SI mandates; same-day reject reporting with reason codes.
Digital KYC acceptance	Acceptance of NADRA-verified digital onboarding under tiered Asaan-style limits — no branch visit for a small member.
Deposit-mobilisation agreement	~1.25%/yr on average daily balances, paid monthly, no clawback on churn, annual review only.
Profit on escrow balances	Idle escrow earns the bank's declared Islamic deposit rate before fund placement; declared monthly.
Intraday liquidity backstop	Small committed intraday facility to bridge redemption-to-payout timing gaps; never overnight, never member credit.
Bank Shariah board sign-off	The partner's own Shariah committee certifies the escrow and placement structure.
Agent network access	Cash-in/out for unbanked members via 1Bill presentment at agent counters.
Dispute and error resolution	Written procedure with timelines and provisional-credit rules for mispostings, duplicates, failed payouts.
Data-sharing and confidentiality	Bank sees aggregate program data, never member-level behavioural IP; we receive account-status webhooks.
Non-exclusivity	Multi-bank future preserved; a second escrow bank as concentration hedge.
Uptime SLA + relationship contact	API availability commitment with maintenance windows; a named human for settlement exceptions.
Co-branded trust assets	Permission to state "collections held in escrow at [Bank]" in-application and in material.

From other institutions and regulators

Asset management LICENCE

ITEM	PURPOSE
Institutional share class, zero loads	No front/back load; management fee as published; trail contracted.
T+0/T+1 redemption, published cutoff	Money the treasury calendar can rely on.
Order + NAV API (or CDC Emlaak)	Machine purchase/redemption; daily rates in-app.
Daily unit-holding statements	Nightly reconciliation to the per-circle registry.
Capacity + gold T+3 terms	Absorption comfort; gold liquidity stated.

Identity and screening LICENCE

ITEM	PURPOSE
NADRA Verisys / BioVerisys	CNIC and biometric verification.
KYC/AML SaaS (Shufti Pro)	Document, face, sanctions and PEP screening.

Credit bureaus LICENCE

ITEM	PURPOSE
eCIB reporting membership	Monthly batch reporting in SBP layout.
DataCheck + Tasdeeq	Scoring and API inquiry; Halqa attributed as furnisher.

Insurance and communications LICENCE PARTNER

ITEM	PURPOSE
Group credit takaful treaty	Auto-enrolment to a free-cover limit; 15–20% distribution share; operator Shariah certificate.
WhatsApp BSP (Infobip / Gupshup)	Templated reminders, invites, receipts.
SMS mask "HALQA"	Branded sender for OTPs and feature-phone nudges.

Regulatory and assurance LICENCE

ITEM	PURPOSE
SECP NBFC licence	Authorisation for arranging and servicing; Oraan precedent.
SECP Regulatory Sandbox	Supervised custody pilot before full licensing.
Retained Shariah advisor + annual audit	Certification of structures, fees, penalty routing, labelling.
Statutory auditor (QCR-rated)	Corporate and client-money assurance.

Production infrastructure LICENCE

ITEM	PURPOSE
Pakistan-resident hosting; managed Postgres (PITR)	Data residency; durable ledger.
KMS/HSM; Cloudflare WAF; Sentry/Grafana	Key custody; DDoS; observability.
External pen-test firm	The report every vendor-risk team demands.

Undertakings we bring — the obligations we hold ourselves to LICENCE

Cybersecurity

- ISO 27001-aligned information-security management system.
- TLS 1.2+ in transit; AES-256 at rest on data and backups.
- KMS/HSM-held keys with rotation; role-based access with admin MFA.
- Secure SDLC: mandatory review, dependency scanning, CI-blocking checks.
- Annual penetration test with evidenced remediation.
- Incident-response plan with regulator and member notification clocks.

Record-keeping and continuity

- Append-only ledger with write-once (WORM) backup tier.
- Retention: 10 years books (Companies Act 2017); 5+ years CDD (AML Act 2010).
- Point-in-time-recovery backups with tested restores.
- Disaster recovery: RPO ≤ 15 minutes, RTO ≤ 4 hours; annual failover drill.
- Data classification and privacy program (PECA 2016 aligned).

Anti-money-laundering (FMU reporting entity)

- Board-approved AML/CFT program under the AML Act 2010.
- Appointed Compliance Officer / MLRO.
- Tiered KYC / CDD / EDD by member value and role.
- Screening against UNSC and NACTA proscribed lists plus PEPs.
- Transaction monitoring calibrated to committee behaviour (velocity, structuring, payer-mismatch) to avoid false-positive flooding.
- STR/CTR filing to the FMU via goAML within prescribed timelines.
- Annual AML training and independent AML audit.

Governance and financial

- Fit-and-proper sponsors and directors; SECP clearance files.
- Minimum equity capital per the licence category.
- Separated compliance and internal-audit functions.
- Consumer-protection framework: plain disclosure, complaint handling, grievance officer, pre-consent fee display.
- Statutory audit plus specific client-money segregation assurance.

Precisely: what works today, and what is a built panel awaiting deployment

Today the platform **records and computes** — it does not hold or move money. The interfaces, engines and logic for the money side are fully built and tested; making them live is the partnership, not more engineering.

Fully functional today BUILT

Works end to end, no external account required — members transact outside, Halqa records.

- **Ledger & audit:** double-entry paisa ledger, append-only audit trail
- **Lifecycle:** create, invite, join, record payment, collect, complete, roll over
- **Conductor:** round scheduling, turn ordering, deadline enforcement, penalties, holdback logic, recovery waterfall bookkeeping, rehabilitation
- **Scoring:** reliability score, credit-weighted / ballot / host-assigned ordering
- **Turn marketplace:** list / bid / buy, premium bounds, fee recorded on the ledger
- **Credit Passport** export (signed, shareable); host reputation card
- **Engine figures** computed and displayed as **indicative** (no placement)
- **Experience:** in-committee chat, Rafa assistant, Urdu RTL, notifications, PWA + native builds
- **Proof:** 120k-agent simulation, 271 integration tests, property tests, CI, scoped error boundaries

Panel + logic built — live money on partnership LICENCE+BANK

The screen, the maths and the rules exist and are tested; what is missing is the account, the custody and the real transfer.

- **Vault device:** allocation sizing, blended-rate maths, projections, risk model, money map are built — **but there are no real balances or fund accounts**; actual holding and real yield need escrow + AMC
- **Earning engines:** float, deposit and allocation are **computed as indicative** today; real fund placement earning actual profit needs escrow + AMC
- **Investment terminal:** dated-rate catalogue and cadence-to-tenor filter built; **actual buying/redeeming** needs the AMC order API
- **Treasury logic:** netting, liquidity ladder, buffers, reconciliation design built; **live escrow custody and payout disbursement** need the bank + rails
- **Auto-cover:** logic built; real settlement from a real vault balance needs custody
- **Payouts & collections:** recorded as references today; **real movement** via Raast / 1Link / virtual IBAN needs the bank
- **Identity & credit rail:** bank-KYC panel (CNIC + IBAN) built; **live NADRA verification and eCIB / bureau reporting** need the licence

What is built, and what activates on partnership

Operational today **BUILT**

- Full committee platform: lifecycle, ledger, scoring, ordering.
- The composable earning engine — all five families of levers, host-configurable.
- Complete default-protection stack: admission gates, graduated collateral, in-cycle enforcement, post-default recovery.
- Vault and investment terminal: allocation, projections, risk model.
- Turn market with atomic settlement.
- Credit passport and host reputation records.
- Urdu interface, assistant, in-committee chat, real-time channel, mobile builds.
- 120,000-agent validation simulation; 271-assertion test suite; continuous integration.

Activates on partnership **LICENCE+BANK**

- Escrow custody and virtual IBAN collections.
- Live fund placement: float, deposits, allocation, vault sleeves.
- Auto-debit mandates and Raast Request-to-Pay.
- Deposit-mobilisation, receivables, trail, takaful, and data lines.
- Bureau reporting of repayment histories.

The engineering is complete and tested; the partnership activates deployment. Integration, not construction.